# Appendix

Anonymous submission

Table S1: Component effect of the proposed dense spatial-temporal modeling.

| | | | | |
|---|---|---|---|---|
| Spatial self-attention | ✓ | ✓ | × | x |
| Temporal self-attention | ✓ | x | ✓ | x |
| BlinkAP$_{0.5:0.95}$ | **24.65** | 18.15 | 20.38 | 12.93 |

# A Experiment

## A.1 DeFB implementation details

DeFB consists of two modules: the face modeling module and the eyeblink modeling module. The face modeling module is trained first. It is responsible for instance-level face localization and obtaining stable eye features and shared queries for the eyeblink modeling module to detect eyeblinks. After the face modeling module is trained, it will perform inference on the entire dataset to obtain the required features. Then, the eyeblink modeling module trains for eyeblink detection based on these features.

For the training of the face modeling module, we initialize the encoder with RT-DETRv2-M (Lv et al. 2024) weights pre-trained on the COCO dataset (Lin et al. 2014) to accelerate convergence and improve generalization. The input video resolution is set to 640×360, the clip length is set to 42. The number of queries $N$ is set to 100, and the number of decoder layers is set to 6. We use the AdamW optimizer with a backbone learning rate of $1 \times 10^{-5}$ and a learning rate of $1 \times 10^{-4}$ for other parts. The warm-up step is set to 2000, with an initial learning rate of 0.1 times the configured learning rate, training for 2 epochs. Training is performed on two 3090 GPUs with a batch size of 8.

For the training of the eyeblink modeling module, the number of sampling points per scale in ROI is set to 5×4. The number of encoding layers in the Dense spatial-temporal modeling (equivalent to performing self-attention operations on eye features alternately in the temporal and spatial dimensions $x$ times) is set to 6, and the number of decoding layers in the Blink decoder is also set to 6. We use the AdamW optimizer with a learning rate of $1 \times 10^{-4}$, training for 20 epochs.

Although the training is divided into two processes, training the face modeling module and the eyeblink modeling module successively, during inference, the two modules of DeFB can be seamlessly integrated for inference, achieving an inference efficiency similar to that of a unified model. During inference, the clip length is set to 42 with a stride of 14. The face localization module and the eyeblink module work together to simultaneously predict tracking results and eyeblink results. The IoU is calculated based on the overlapping parts of the tracking results, and the final output is obtained by combining the tracking and eyeblink results.

## A.2 Baseline implementation on MPEblink

For multi-stage methods, we follow the training and testing protocols used in prior work (Zeng et al. 2023) on MPEblink. Specifically, we first use SOTA face detector (Guo et al. 2022) to detect the facial locations or landmarks in each frame. By calculating feature similarity, we link the results across frames to obtain the complete locations of eyeblink instances or eye keypoints. After cropping the eye regions or acquiring the facial landmarks, we input them into the eyeblink detector for eyeblink detection, which is the standard procedure for multi-stage eyeblink detection in multi-person eyeblink videos. For the unified method, we directly adopt InstBlink using its open-source code, following standard practices.

## A.3 Ablation on dense spatio-temporal modeling

The decoupled feature strategy in DeFB refines coarse features through joint spatial (cross-pixel) and temporal (cross-frame) modeling, effectively enhancing fine-grained appearance and motion modeling for eyes, which significantly improves eyeblink detection accuracy. The core idea stems from our decomposed framework: although eye modeling builds on face features, it is isolated via a dedicated module with asynchronous training to enable stable and focused learning. The specific architecture is not our main focus (even simple self-attention proves effective as shown in Tab. S1) and can be replaced with any stronger alternatives.

## A.4 DeFB as plug-in: implementation details

The baseline implementation adopts a framework similar to two-stage eyeblink detection methods. It first employs a SOTA face detector (Guo et al. 2022) to localize eye regions, then extracts these ocular regions as cropped images to train a general action detector for eyeblink detection task.

When DeFB functions as a plug-in, the action detector essentially replaces the eyeblink modeling module in the

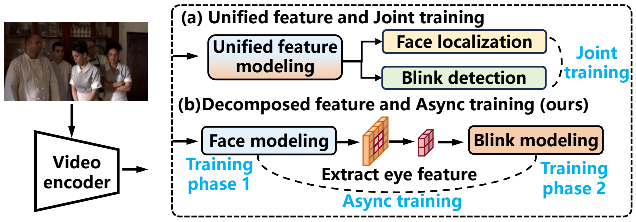


Figure S1: Comparison of training paradigms: (a) SOTA method's unified feature modeling with joint training; (b) Our decomposed feature modeling with async training.

original DeFB architecture. The training process follows a staged strategy consistent with DeFB's core design: the facial modeling module is trained independently first. Once converged, the pre-trained facial modeling module is used to extract eye-region features and shared queries. These features and queries then serve as inputs to train the action detector, enabling it to perform eyeblink detection within the decomposed feature learning paradigm. This integration retains DeFB's advantages in stable face feature learning while leveraging the action detector's architecture for eyeblink detection.

## B Comparison of training paradigms between DeFB and InstBlink

The differences in training paradigms between DeFB and InstBlink are illustrated in Fig . S1, which can be summarized as follows: (1) **Feature Space Strategy**: InstBlink adopts a unified feature space where both face localization and eyeblink detection tasks share the same query, which interacts with video features of a single granularity for prediction. In contrast, DeFB employs decomposed feature spaces. While retaining a shared query for efficient feature sharing across face localization and eyeblink detection tasks, DeFB introduces a two-stage interaction mechanism: the query first interacts with coarsely modeled video features to predict face positions. Subsequently, based on the predicted face positions and coarse-grained features, it refines and extracts fine-grained features specific to the eye region. The query then interacts with these fine-grained eye features for eyeblink detection. This decomposed strategy leverages coarser global features for face modeling to enhance speed, while utilizing finer local eye features to improve detection accuracy. (2) **Training Mechanism**: InstBlink utilizes a multi-task learning paradigm that jointly optimizes both face localization and eyeblink detection tasks. DeFB, however, adopts an asynchronous training paradigm, where the facial modeling module and eyeblink modeling module are trained sequentially. This approach avoids interference caused by unstable face-eye features—particularly those arising from immature facial representations during the early stages of joint training—thereby stabilizing the learning process for eyeblink detection.

## References

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